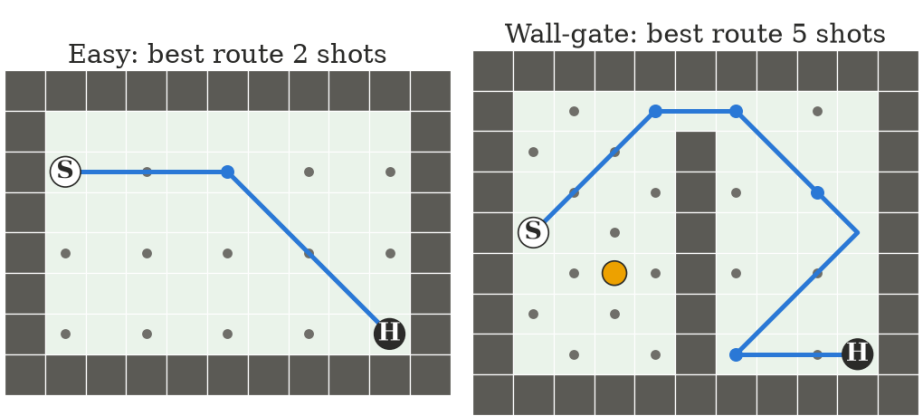
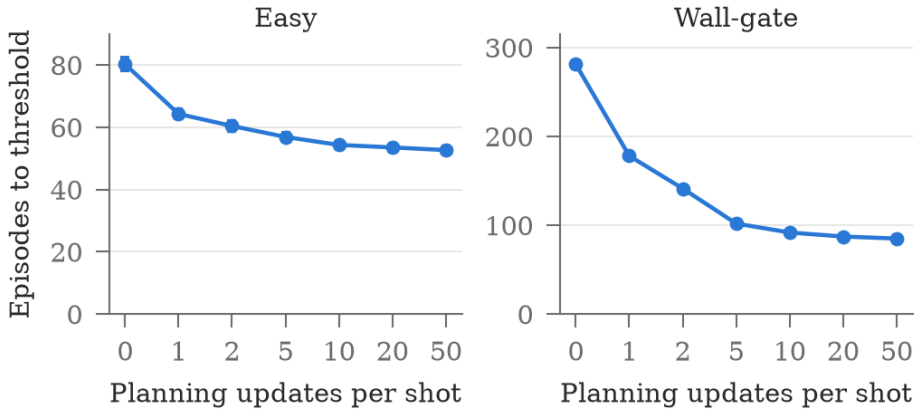
# More imagined practice means fewer real shots




Q-learning updates once per real shot.

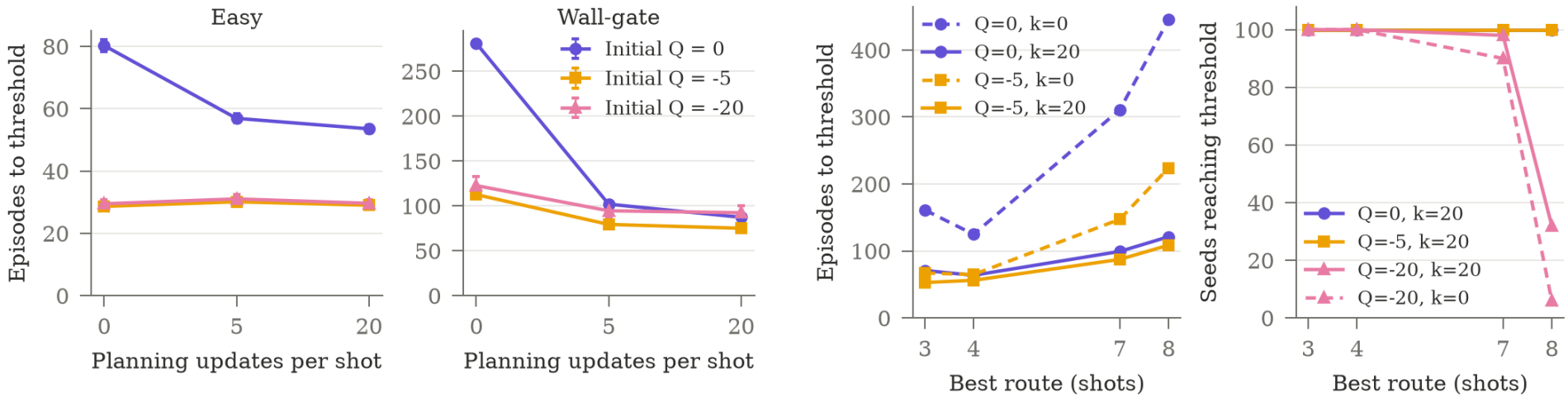
Dyna-Q also replays k remembered shots after each real one.

Wall-gate real shots to 80% success: 5063, 1848, 1607 for k = 0, 5, 20 (5 seeds).

Most of the gain comes from the first few updates (100 seeds, right).

Threshold: first 20-episode window with at least 80% sinks. Bars show 95% intervals.

# But it depends on where the values start




Every shot costs 1, so a zero Q table is optimistic. Planning partly drains that.

Without optimism the benefit disappears on the 2-shot course and shrinks on Wall-gate.

With the lowest start value, most agents never solve the 8-shot course.

A fair baseline for a planner is a well-initialized learner.

100 seeds, separate from the primary five. Exploratory: designed after the first results.